string.py

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# 1. Write a program to input a string and display its length without using the len() function.
# x = input("Enter a string: ")
# count = 0
# for ch in x:
#     count += 1
# print("Length of the string is:", count)

# 2.Count the number of vowels, consonants, digits, spaces, and special characters in a given string.
# text = input("Enter a string: ")
# vowels = consonants = digits = spaces = special = 0
# for ch in text:
#     if ch.isalpha():
#         if ch.lower() in "aeiou":
#             vowels += 1
#         else:
#             consonants += 1
#     elif ch.isdigit():
#         digits += 1
#     elif ch.isspace():
#         spaces += 1
#     else:
#         special += 1
# print("Vowels:", vowels)
# print("Consonants:", consonants)
# print("Digits:", digits)
# print("Spaces:", spaces)
# print("Special Characters:", special)

# 3.Reverse the given string without using built-in reverse functions.
# text = input("Enter a string: ")
# reverse = ""
# for ch in text:
#     reverse = ch + reverse
# print("Reversed string:", reverse)


# 4.Check whether the entered string is a palindrome.
# Input string
# text = input("Enter a string: ")
# reverse = ""
# for ch in text:
#     reverse = ch + reverse
# if text == reverse:
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#     print("Palindrome")
# else:
#     print("Not a palindrome")


# 5.Count the number of uppercase and lowercase letters in a string.
# Input string
# text = input("Enter a string: ")
# uppercase = 0
# lowercase = 0
# for ch in text:
#     if ch.isupper():
#         uppercase += 1
#     elif ch.islower():
#         lowercase += 1
# print("Uppercase letters:", uppercase)
# print("Lowercase letters:", lowercase)


# 6.Replace all occurrences of a given character with another character.
# text = input("Enter a string: ")
# old = input("Enter the character to replace: ")
# new = input("Enter the new character: ")
# result = ""
# for ch in text:
#     if ch == old:
#         result += new
#     else:
#         result += ch
# print("Updated string:", result)

# 7.Remove all spaces from the input string.
# text = input("Enter a string: ")
# result = ""
# for ch in text:
#     if ch != " ":
#         # result += ch
# print("String without spaces:", result)

# 8.Find the number of times a specified character appears in a string.
# text = input("Enter a string: ")
# char = input("Enter the character to count: ")
# count = 0
# for ch in text:
#     if ch == char:
#         count += 1
# print("Number of occurrences:", count)
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# 9.Print the first and last character of a string.
# text = input("Enter a string: ")
# print("First character:", text[0])
# print("Last character:", text[-1])

# 10.Display each character of a string along with its ASCII value.
# text = input("Enter a string: ")
# for ch in text:
#     print(ch, ":", ord(ch))

# 11.Count the total number of words in a sentence.
# sentence = input("Enter a sentence: ")
# count = 0
# for word in sentence.split():
#     count += 1
# print("Total number of words:", count)

# 12.Find the longest word in a given sentence.
# sentence = input("Enter a sentence: ")
# words = sentence.split()
# longest = ""
# for word in words:
#     if len(word) > len(longest):
#         longest = word
# print("Longest word:", longest)

# 13.Find the shortest word in a sentence.
# sentence = input("Enter a sentence: ")
# words = sentence.split()
# shortest = words[0]
# for word in words:
#     if len(word) < len(shortest):
#         shortest = word
# print("Shortest word:", shortest)

# 14.Convert the first letter of every word to uppercase.
# sentence = input("Enter a sentence: ")
# result = ""
# for word in sentence.split():
#     result += word[0].upper() + word[1:] + " "
# print("Updated sentence:", result)

# 15.Print all duplicate characters in a string.
# text = input("Enter a string: ")
# duplicates = ""
# for ch in text:
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#     if text.count(ch) > 1 and ch not in duplicates:
#         duplicates += ch
# print("Duplicate characters:", duplicates)


# 16.Display the frequency of every character in a string
# text = input("Enter a string: ")
# frequency = {}
# for ch in text:
#     if ch in frequency:
#         frequency[ch] += 1
#     else:
#         frequency[ch] = 1
# print("Character frequency:")
# for ch, count in frequency.items():
#     print(ch, ":", count)

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